

Trinadh Musunuri

trinadh.musunuri@gmail.com | linkedin.com/in/trinadh-musunuri | github.com/3nadh3 | trinadh.dev

EDUCATION

Central Michigan University

Master of Science in Computer Science

Mount Pleasant, MI

Expected May 2027

- **College:** College of Science and Engineering
- **Relevant Coursework:** Advanced Algorithms, Cloud Computing, Web Technologies, Machine Learning

EXPERIENCE

IBM

Software Engineer Intern, watsonx Orchestrate

Austin, TX

May 2026 – Present

- Engineered a multi-agent AI platform using React and IBM watsonx Orchestrate, integrating Slack delivery with 3 structured widget types (Knowledge Quiz, Weather Cards, Score Cards) for enterprise workflows.
- Designed and implemented a Voice Call Metrics / CDR pipeline spanning **wxo-server** and **ai-gateway**, enabling end-to-end call trace collection and observability for voice agents for Testing and Debugging.

Central Michigan University

Research Assistant

Mount Pleasant, MI

Sep 2025 – May 2026

- Replaced an evolutionary algorithm with a gradient-based optimization method, achieving **50× faster** runtime (30–60 min → under 3 min) on adversarial ML experiments.
- Conducted low-level architectural analysis on AMD Ryzen AI hardware (XDNA 2 / Krackan), applying fine-grained cycle counting and DRAM contention metrics to profile NPU execution patterns.
- Designed and executed CPU–NPU benchmarking scripts to collect high-fidelity performance data across deep learning workloads, contributing directly to hardware security research publications.
- Engineered an interactive adversarial YOLOv2 attack demo—currently in active use by students—hosted on Hugging Face Spaces with a Cloudflare Worker proxy to circumvent restrictive university firewalls.

PROJECTS

CyberGuard XAI — Phishing Detection with Explainable AI

- Built a phishing email classifier using DistilBERT with LIME and SHAP explainability, exposing per-word risk-drop scores so users can interpret and challenge model decisions.
- Developed an interactive evasion guide featuring word swap/undo, paginated risk cards, and a live contentEditable editor with real-time LIME score tooltips for adversarial text exploration.
- Deployed React/Vite frontend to Netlify and FastAPI backend to Hugging Face Spaces and Render with a fully automated CI/CD pipeline.

SkillSwap — Full Stack Skill Matching & Messaging Platform

- Built a full-stack peer skill exchange platform using the MERN stack with WebSocket-based real-time messaging and JWT-secured REST APIs.
- Deployed on cloud infrastructure with environment-secured configuration, MongoDB data models, and scalable backend architecture.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, SQL

Web & APIs: React.js, Node.js, Express.js, FastAPI, Vite, REST, WebSockets, JWT

AI & Machine Learning: PyTorch, IBM watsonx, Vertex AI, DistilBERT, YOLOv2, LIME, SHAP

Cloud & DevOps: AWS (EC2, S3, IAM, RDS, Lambda), Google Cloud, IBM Cloud, Docker, CI/CD, Netlify, Render

Databases: MongoDB, MySQL

Tools: Git, GitHub, Postman, Hugging Face, OPENAPI

CERTIFICATIONS

Generative AI Engineering — Google Cloud

- Vertex AI, prompt engineering, model deployment, scalable ML pipelines, cloud-based AI workflows